3 pct prezență

7 pct. ieșit la tablă

Șiruri de nr. reale - recapitulare

1) $\lim\limits_{n\to\infty} \sqrt[3]{n}\left(\sqrt[3]{(n+1)^2}-\sqrt[3]{(n-1)^2}\right)$

2) $\lim\limits_{n\to\infty} \cos\left(n\pi\sqrt[2n]{e}\right)$

3) $\lim\limits_{n\to\infty}\left(1-\ln\frac{2}{1}+\frac{1}{2}-\ln\frac{3}{2}+\dots+\frac{1}{n}-\ln\frac{n+1}{n}\right)$

4) $\lim\limits_{n\to\infty}\frac{n!\,2^n}{(n+1)(n+2)\dots(2n)}$

5) $\lim\limits_{n\to\infty}\frac{1+2\sqrt{2}+3\sqrt{3}+\dots+n\sqrt{n}}{n^2\sqrt{n}}$

6) $\lim\limits_{n\to\infty}\sqrt[n]{\frac{n^{2n+1}}{(2n)!}}$

7) $\lim\limits_{n\to\infty}\frac{1\cdot3\cdot5\cdot\ldots\cdot(2n-1)}{2\cdot4\cdot6\cdot\ldots\cdot(2n)}$

8) $\lim\limits_{n\to\infty}\sqrt{n}\left[\left(1+\frac{1}{n}\right)^{n+1}-\left(1+\frac{1}{n}\right)^{n}\right]$

1) $$\lim_{n\to\infty}\sqrt[3]{n}\,\frac{(n+1)^2-(n-1)^2}{\sqrt[3]{(n+1)^4}+\sqrt[3]{(n+1)^2(n-1)^2}+\sqrt[3]{(n-1)^4}}\ \underset{=}{\overset{Not}{=}}\ L_1$$

$$L_1=\lim_{n\to\infty}\sqrt[3]{n}\,\frac{\cancel{n^2}+2n+\cancel{1}-\cancel{n^2}+2n-\cancel{1}}{\sqrt[3]{(n+1)^4}+\sqrt[3]{(n-1)^4}+\sqrt[3]{(n^2-1)^2}}$$

$$L_1=\lim_{n\to\infty}\frac{4\cancel{n\sqrt[3]{n}}}{\cancel{n\sqrt[3]{n}}\left(\sqrt[3]{\left(1+\frac{1}{n}\right)^4}+\sqrt[3]{\left(1-\frac{1}{n}\right)^4}+\sqrt[3]{\left(1-\frac{1}{n^2}\right)^2}\right)}$$

$$L_1=\frac{4}{(1+0)+(1-0)+(1-0)}=\frac{4}{3}$$

3) $$\lim_{n\to\infty}\left[1+\frac{1}{2}+\dots+\frac{1}{n}-\ln\left(\frac{\cancel{2}}{1}\cdot\frac{\cancel{3}}{\cancel{2}}\cdot\frac{\cancel{4}}{\cancel{3}}\cdot\ldots\cdot\frac{\cancel{n}}{\cancel{n-1}}\cdot\frac{n+1}{\cancel{n}}\right)\right]$$

$$L_3=\lim_{n\to\infty}\left[1+\frac{1}{2}+\dots+\frac{1}{n}-\ln(n+1)\right]$$

fie $\gamma_n=1+\frac{1}{2}+\dots+\frac{1}{n}-\ln(n)$

arătăm că γ_n e convergent

$$\gamma_{n+1}-\gamma_n=\cancel{1+\frac{1}{2}+\dots}+\frac{1}{n}-\ln(n)\cancel{-1-\frac{1}{2}-\dots}-\frac{1}{n}+\frac{1}{n+1}+\ln(n+1)$$
$$=\ln\left(\frac{n+1}{n}\right)+\frac{1}{n+1}$$

$a_n=\left(1+\frac{1}{n}\right)^n$ arătăm că a_n e strict. cresc.

$$a_n=C_n^0 1+C_n^1\frac{1}{n}+C_n^2\frac{1}{n^2}+C_n^3\frac{1}{n^3}+\dots+C_n^n\frac{1}{n^n}$$

$$a_n=1+n\cdot\frac{1}{n}+\frac{n(n-1)}{2!}\cdot\frac{1}{n^2}+\frac{n(n-1)(n-2)}{3!}\cdot\frac{1}{n^3}+\dots+\frac{1}{n^n}$$

$$a_n=2+\left(1-\frac{1}{n}\right)\cdot\frac{1}{2!}+\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\cdot\frac{1}{3!}+\dots+\frac{1}{n!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{n-1}{n}\right)$$

$$a_n=2+\sum_{k=1}^{n}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\dots\left(1-\frac{k-1}{n}\right)\cdot\frac{1}{k!}$$

$$a_{n+1}=2+\sum_{k=2}^{n+1}\left(1-\frac{1}{n+1}\right)\left(1-\frac{2}{n+1}\right)\dots\left(1-\frac{k}{n+1}\right)\frac{1}{k!}$$

$\Rightarrow a_n<a_{n+1}$ prin comp. de fiecare termen

$\lim\limits_{n\to\infty}a_n=e$

$b_n=\left(1+\frac{1}{n}\right)^{n+1}$ arătăm că e strict desc.

$$\frac{b_n}{b_{n+1}}=\frac{\left(1+\frac{1}{n}\right)^{n+1}}{\left(1+\frac{1}{n+1}\right)^{n+2}}=\frac{(n+1)^{n+1}}{n^{n+1}}\cdot\frac{(n+1)^{n+2}}{(n+1+1)^{n+2}}=\frac{(n+1)^{2n+3}}{n^{n+1}\cdot(n+2)^{n+2}}$$

$$\frac{b_n}{b_{n+1}}=\frac{\left[(n+1)^2\right]^{n+1}\cdot(n+1)}{\left[n(n+2)\right]^{n+1}\cdot(n+2)}=\left(\frac{n^2+2n+1}{n^2+2n}\right)^{n+1}\cdot\frac{n+1}{n+2}$$

$$\frac{b_n}{b_{n+1}}=\left(1+\frac{1}{n^2+2n}\right)^{n+1}\cdot\frac{n+1}{n+2}$$

$\lim\limits_{n\to\infty}b_n=e$

Inegalitatea lui Bernoulli

$(1+x)^n\geq 1+nx \quad \forall x\in\mathbb{R},\ x>-1$
$\forall n\in\mathbb{N}^*$

$$\frac{b_n}{b_{n+1}}\geq\frac{n+1}{n+2}\left[1+(n+1)\frac{1}{n^2+2n}\right]=\frac{n+1}{n+2}\cdot\frac{n^2+3n+1}{n^2+2n}$$

$$\frac{b_n}{b_{n+1}}\geq\frac{n^3+3n^2+n+n^2+3n+1}{n^3+2n^2+2n^2+4n}=\frac{n^3+4n^2+4n+1}{n^3+4n^2+4n}>1$$

$\Rightarrow b_n>b_{n+1}\Rightarrow b_n\downarrow$

$$\left(1+\frac{1}{n}\right)^n<e<\left(1+\frac{1}{n}\right)^{n+1}$$

$$n\cdot\ln\left(1+\frac{1}{n}\right)<1<(n+1)\ln\left(1+\frac{1}{n}\right)$$

$$n\ln\left(\frac{n+1}{n}\right)<1<(n+1)\ln\frac{n+1}{n}\Rightarrow\frac{1}{n+1}<\ln\frac{n+1}{n}<\frac{1}{n}$$

$$\gamma_{n+1}-\gamma_n=\frac{1}{n+1}-\ln\frac{n+1}{n}<0\Rightarrow\gamma_n\downarrow\ (I)$$

$\ln\frac{2}{1}<1$

$\ln\frac{3}{2}<\frac{1}{2}$

$\vdots$

$\ln\frac{n}{n-1}<\frac{1}{n-1}$

$\ln\frac{n+1}{n}<\frac{1}{n}$

$$\Rightarrow\ln\left(\frac{2}{1}\cdot\frac{3}{2}\cdot\ldots\cdot\frac{n+1}{n}\right)<1+\frac{1}{2}+\dots+\frac{1}{n}$$

$$\ln(n+1)-\ln(n)<\gamma_n$$

$$\underset{0}{\ln\left(\frac{n+1}{n}\right)}<\gamma_n\ \not\Rightarrow\ \gamma_n\in(0;1)$$

$$(I);(II)\ \xRightarrow{\text{Weierstr.}}\ \gamma_n \text{ e convergent}\Rightarrow\ (II)$$
$$\Rightarrow\lim_{n\to\infty}\gamma_n=C$$

$$L_3=\lim_{n\to\infty}\left[1+\frac{1}{2}+\dots+\frac{1}{n}-\ln(n+1)\right]=$$

$$L_3=\lim_{n\to\infty}\left[\underbrace{1+\frac{1}{2}+\dots+\frac{1}{n}-\ln(n)}_{C}+\ln(n)-\ln(n+1)\right]$$

$$\frac{1}{n+1}<\ln\frac{n+1}{n}<\frac{1}{n}$$

$$\frac{1}{n+1}<-\ln\frac{n}{n+1}<\frac{1}{n}$$

$$\lim_{n\to\infty}\left[\ln(n)-\ln(n+1)\right]=0$$

$L_3=C-0\Rightarrow L_3=C$

5) $$\lim_{n\to\infty}\frac{1+2\sqrt{2}+3\sqrt{3}+\dots+n\sqrt{n}}{n^2\sqrt{n}}=\lim_{n\to\infty}\frac{a_n}{b_n}$$

$$\lim_{n\to\infty}\frac{a_{n+1}-a_n}{b_{n+1}-b_n}=\lim_{n\to\infty}\frac{(n+1)\sqrt{n+1}}{(n+1)^2\sqrt{n+1}-n^2\sqrt{n}}=$$

$$=\lim_{n\to\infty}\frac{(n+1)\sqrt{n+1}}{\sqrt{(n+1)^5}-\sqrt{n^5}}=\lim_{n\to\infty}\frac{(n+1)\sqrt{n+1}\left(\sqrt{(n+1)^5}+\sqrt{n^5}\right)}{(n+1)^5-n^5}$$

$$=\lim_{n\to\infty}\frac{\sqrt{(n+1)^3}\left(\sqrt{(n+1)^5}+\sqrt{n^5}\right)}{\cancel{n^5}+C_5^1n^4+C_5^2n^3+C_5^3n^2+C_5^4n+C_5^5-\cancel{n^5}}=$$

$$=\lim_{n\to\infty}\frac{n^4\sqrt{\left(1+\frac{1}{n}\right)^3}\left(\sqrt{\left(1+\frac{1}{n}\right)^5}+1\right)}{5n^4+C_5^2n^3+C_5^3n^2+C_5^4n+1}=\frac{1\cdot2}{5}=\frac{2}{5}$$

$$\xRightarrow[\text{Stolz Cesaro}]{\text{cf.}}\lim_{n\to\infty}\frac{a_n}{b_n}=L_5=\frac{2}{5}$$

7) $$\lim_{n\to\infty}\frac{1\cdot3\cdot5\cdot\ldots\cdot(2n-1)}{2\cdot4\cdot6\cdot\ldots\cdot(2n)}=\lim_{n\to\infty}x_n=L_7$$

$\frac{1}{2}\leq\frac{1}{2}<\frac{2}{3}$

$\frac{2}{3}<\frac{3}{4}<\frac{4}{5}$

$\frac{4}{5}<\frac{5}{6}<\frac{6}{7}$

$\vdots$

$\frac{2n-2}{2n-1}<\frac{2n-1}{2n}<\frac{2n}{2n+1}$

$$\Rightarrow\frac{1}{2}\cdot\frac{2}{3}\cdot\ldots\cdot\frac{2n-2}{2n-1}<\frac{1}{2}\cdot\frac{3}{4}\cdot\ldots\cdot\frac{2n-1}{2n}<\frac{2}{3}\cdot\frac{4}{5}\cdot\ldots\cdot\frac{2n}{2n+1}$$

$$\frac{1}{2}\cdot\frac{1}{2n}<x_n^2<\frac{1}{2n+1}$$

$$\frac{1}{2\sqrt{n}}<x_n<\frac{1}{\sqrt{2n+1}}$$

$$\Rightarrow\lim_{n\to\infty}x_n=0$$

8) $$\lim_{n\to\infty}\sqrt{n}\left[\left(1+\frac{1}{n}\right)^{n+1}-\left(1+\frac{1}{n}\right)^{n}\right]=\lim_{n\to\infty}\sqrt{n}\left(1+\frac{1}{n}\right)^n\left(1+\frac{1}{n}-1\right)$$

$$=\lim_{n\to\infty}\sqrt{n}\cdot\frac{1}{n}\left(1+\frac{1}{n}\right)^n=0\cdot e=0$$

Criteriul raportului

$x_n>0\quad\forall n$ Dacă $l<1\Rightarrow x_n\xrightarrow[n\to\infty]{}0$

$\lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}=l$ Dacă $l>1\Rightarrow x_n\xrightarrow[n\to\infty]{}\infty$

4) $$\lim_{n\to\infty}\frac{n!\,2^n}{(n+1)(n+2)\cdot\ldots\cdot(2n)}=\lim_{n\to\infty}x_n$$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(n+1)!\cdot2^{n+1}}{(n+2)\cancel{(n+3)\dots}(2n+1)(2n+2)}\cdot\frac{(n+1)\cancel{\dots(2n)}}{n!\,2^n}$$

$$=\lim_{n\to\infty}\frac{2\cdot(n+1)^2}{(2n+1)(2n+2)}=\lim_{n\to\infty}\frac{2(n^2+2n+1)}{4n^2+4n+2n+2}=\frac{1}{2}<1$$

$$\Rightarrow\lim_{n\to\infty}x_n=0$$

Consecință Stolz-Cesaro

$x_n>0$ și $l=\lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}$ Atunci $\lim\limits_{n\to\infty}\sqrt[n]{x_n}=l$

6) $$\lim_{n\to\infty}\sqrt[n]{\frac{n^{2n+1}}{(2n)!}}=\lim_{n\to\infty}\sqrt[n]{x_n}$$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(n+1)^{2n+3}}{(2n+2)!}\cdot\frac{(2n)!}{n^{2n+1}}=\lim_{n\to\infty}\frac{\overset{(n+1)^{2n+2}}{\cancel{(n+1)^{2n+3}}}}{n^{2n+1}(2n+1)\underset{2}{\cancel{(2n+2)}}}$$

$$=\lim_{n\to\infty}\left(\frac{n+1}{n}\right)^{2n+1}\cdot\frac{n+1}{2(2n+1)}=e^2\cdot\frac{1}{4}=\frac{e^2}{4}$$

$$\left(1+\frac{1}{n}\right)^{2n+1}=\Big[\underbrace{\left(1+\frac{1}{n}\right)^n}_{\to e^2}\Big]^2\cdot\underbrace{\left(1+\frac{1}{n}\right)}_{\to 1}$$

2) $$\lim_{n\to\infty}\cos\left(n\pi\sqrt[2n]{e}\right)=L_2$$

$$\cos\left(n\pi\sqrt[2n]{e}-n\pi+n\pi\right)=\cos\left(n\pi\sqrt[2n]{e}-n\pi\right)\underbrace{\cos(n\pi)}_{(-1)^n}-\sin\left(n\pi\sqrt[2n]{e}-n\pi\right)\underbrace{\sin(n\pi)}_{0}$$

$$L_2=\lim_{n\to\infty}(-1)^n\cos\left[n\pi\left(\sqrt[2n]{e}-1\right)\right]$$

$$L_2=\lim_{n\to\infty}\underbrace{(-1)^n}_{\text{mărginit}}\underbrace{\cos\Bigg[\frac{\pi}{2}\underbrace{\frac{e^{\frac{1}{2n}}-1}{\frac{1}{2n}}}_{1}\Bigg]}_{\cos\frac{\pi}{2}=0}=0$$